Experiment 3: Water Jug Problem

Aim

To solve the Water Jug problem using Python.

Objective

To measure the required amount of water using two jugs of different capacities by performing fill, empty, and transfer operations.

Algorithm

1. Take two jugs with capacities 4 L and 3 L.
2. Fill the 4 L jug.
3. Pour water from the 4 L jug into the 3 L jug.
4. Empty the 3 L jug.
5. Transfer the remaining water from the 4 L jug to the 3 L jug.
6. Fill the 4 L jug again.
7. Pour water into the 3 L jug until it becomes full.
8. The remaining water in the 4 L jug is 2 L.
9. Display the steps and stop.

Flowchart

Start

↓

Fill 4 L Jug

↓

Pour into 3 L Jug

↓

Is 3 L Jug Full?

Yes No

↓

Empty 3 L Jug

↓

Transfer Remaining Water

↓

Fill 4 L Jug Again

↓

Pour into 3 L Jug

↓

Is 2 L Obtained?

Yes No

↓

Display Result

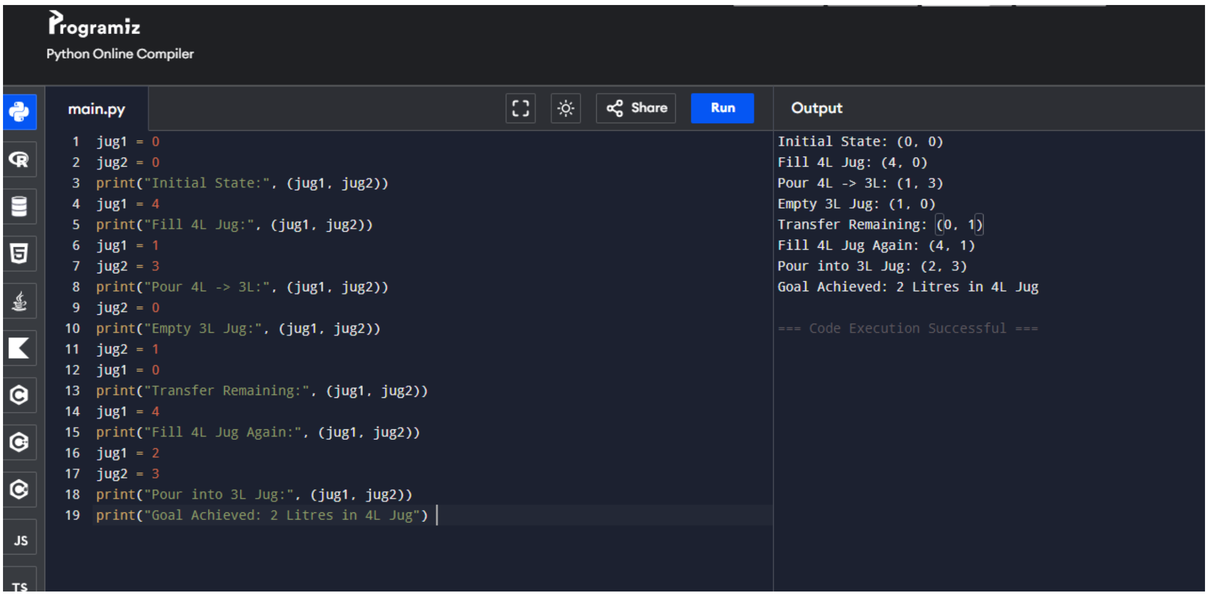
↓

Stop

Python Program

```
jug1 = 0
jug2 = 0

print("Initial State:", (jug1, jug2))

jug1 = 4
print("Fill 4L Jug:", (jug1, jug2))

jug1 = 1
jug2 = 3
print("Pour 4L -> 3L:", (jug1, jug2))

jug2 = 0
print("Empty 3L Jug:", (jug1, jug2))

jug2 = 1
jug1 = 0
print("Transfer Remaining:", (jug1, jug2))

jug1 = 4
print("Fill 4L Jug Again:", (jug1, jug2))

jug1 = 2
jug2 = 3
print("Pour into 3L Jug:", (jug1, jug2))

print("Goal Achieved: 2 Litres in 4L Jug")
```

Sample Output



Result

Thus, the Water Jug problem was solved successfully using Python to obtain **2 litres** of water.

Conclusion

The program demonstrates how the **fill, empty, and transfer operations** can be used to measure the required quantity of water. It successfully obtains **2 litres** in the 4 L jug.